



Program and Accelerator Monitoring, Evaluation, Learning and Impact Assessment (MELIA) Plan

Genebanks Accelerator

June 2025

Table of Contents

1. Program/Accelerator overview and results framework.....	3
Program/Accelerator overview	3
Theory of change	4
Results framework.....	7
2. Monitoring plan for HLOs and outcomes.....	7
Data collection strategy	7
3. Evaluation plan.....	8
Evaluation studies	8
Evaluation collaboration plans	8
4. Impact assessment.....	8
Program/Accelerator impact overview	9
IA activities and collaboration plans	11
IA studies	12
5. Learning and adaptive management	12
Learning.....	12
Annual review and adaptive planning	13
6. MELIA resources.....	14
MELIA budget and staff	14
7. Data management.....	14
General approach.....	14
Annex 1. MELIA studies.....	15
Annex 2. MELIA budget.....	18

1. Program/Accelerator overview and results framework

Program/Accelerator overview

Description

As global temperature increases surpass the 1.5oC threshold, we enter uncharted territory. We will require unprecedented resilience to survive climate challenges and mass biodiversity extinction. As part of its Zero Hunger goal, the UN has set a target for the genetic diversity of crops to be maintained and made accessible from soundly managed genebanks. Under the Plant Treaty, CGIAR has a legal obligation to conserve in the long term and make available the crop diversity in collections managed by 11 Centers in 13 countries. Ten CGIAR genebanks come together in the Genebanks Accelerator to make this target achievable.

Thanks to associated germplasm health units (GHU) and their strategic location in diversity hotspots, Genebanks connect widely with users in low- and middle-income countries, making accessible more than 3,500 plant species to researchers, breeders, extension programs, local communities, schools, and farmers around the world, supporting them to deal with intensifying challenges on local to global scales and to explore a multitude of options for a more resilient, diverse and environmentally sound agriculture and landscape.

Genebanks, through five Areas of Work, will enable diversity conservation in a more efficient and effective manner, smarter and more targeted access to diversity for increased use, and a supportive policy environment while controlling phytosanitary risks. The genebanks will share their expertise and capacity with regional and national partners to connect and integrate collections on a global scale and create a more vibrant and dynamic two-way flow between in situ and ex situ. Frontier technologies and innovations in genomics, phenomics, artificial intelligence (AI), and cryopreservation are enhancing the relevance of genebanks, placing our generation in a position of responsibility to explore and use genetic diversity more effectively and on a much wider scale than previously, so as to reverse the narrowing of the base upon which food security and diets now rests. In 2023, the genebanks distributed over 200,000 germplasm samples to more than 60 countries. Over the next six years, through strengthened partnerships with Breeding for Tomorrow, Digital Transformation, Multifunctional Landscapes, and other users, Genebanks will develop enriched digital resources and harness artificial intelligence and machine learning to deploy the diversity in collections to its full potential. To support this ambition, Genebanks will strengthen open data policies and help build the capacity of potential genebank users to access and use the genetic and data resources available to them. More intensive and widespread use of such resources will lead to diversification in farmers' fields and household gardens. This will strengthen adaptation to climate change, improve nutrition and food security, incomes, environmental health, and support social inclusion, marginalized people, and communities.

By conserving and making available biodiversity, Genebanks directly contributes to CGIAR's Collective Target 11, UN Sustainable Development Goal 2.5, Kunming-Montreal Global Biodiversity Framework Targets 4, 10, 13, and 20, and the FAO's Global Plan of Action on Plant Genetic Resources for Food and Agriculture (PGRFA).

Components

The Genebanks Accelerator will be delivered through five interconnected Areas of Work (AoWs):

AoW 1: Biodiversity Conservation

AoW 2: Strategic User Engagement

AoW 3: Genetic Resources Policy

AoW 4: Germplasm Health

AoW 5: Strengthening Capacity for In Situ and Ex Situ Conservation Globally

Theory of change

Narrative

The full Theory of Change and AoW Theory of Change narratives with documented assumptions are found <https://toc.mel.cgiar.org/toc/16685dd2-9da6-4a10-9270-dfdfb2dcdb53?relatedFlowId=39e86294-d092-4949-bb34-8857b4d5f021>

The Genebanks Accelerator aims to ensure the reliability of CGIAR conservation and genebank services now and into the future and to build a stronger global system of policies, mechanisms, institutions and conservationists making genetic diversity more widely available. The Genebanks Accelerator will be delivered through five interconnected Areas of Work (AoWs): AoW 1. Biodiversity Conservation, AoW 2: Strategic User Engagement, AoW 3: Genetic Resources Policy, AoW 4: Germplasm Health, AoW 5: Strengthening capacity for in situ and ex situ conservation globally.

Through AoW 1, CGIAR Genebanks will conserve and make available crop diversity in perpetuity with greater efficiency and effectiveness. AoW 1 supports the genebanks' operations and improving their related processes, collections, and facilities. CGIAR is uniquely placed to pilot and refine new conservation-related technologies across diverse locations and crops. AoW 1 supports genebanks in reaching and sustaining performance targets so that they become eligible for Long-Term Partnerships with the Crop Trust. AoW 1 will maintain and create new partnerships with ARIs and universities, NARS and national genebanks, the Crop Trust, and the Plant Treaty Secretariat.

AoW 2 will enable researchers, breeders, students and other users to access diversity and associated data in a smarter and more targeted way for increased use. AoW 2 will implement a strategic approach to identify, prioritize, and characterize user typologies, analyze their needs, and proactively engage with user type representatives to determine how to improve the user experience in finding, selecting, and requesting germplasm from CGIAR genebanks. Enriched information, new products, and responsible AI innovation will help users identify materials and improve their experience ordering and obtaining germplasm, which is expected to facilitate breeding and research and potentially radically speed up their research. Improved user tools and outreach will simultaneously create more demand for genetic diversity and make it possible for users to target more precisely what germplasm from CGIAR hosted collections has the most potential to serve their needs. Other trends and external factors will also have a strong influence on germplasm distribution, creating challenges for determining suitable targets for this key

output and creating a risk of setting up counter-productive incentives. Successful user engagement may also generate opportunities to attract funding and sponsorship to support long-term conservation. AoW 2 will work with Breeding for Tomorrow, Digital Transformation, partner with NARS and universities, and establish a user feedback loop.

AoW 3 (Genetic Resources Policy) will a) ensure CGIAR scientists comply with evolving international and national access and benefit sharing as materials move across international borders and b) contribute to ongoing international negotiations to make those laws more supportive of Ag for development, and c) work with national partners to develop complementary national measures implementing those same laws. AoW 3 will focus primarily on the Plant Treaty's multilateral system of access and benefit sharing, but making sure it evolves and is implemented in ways that are in harmony with the Convention on Biological Diversity's Nagoya Protocol on ABS and Decision 16/2 on digital sequence information. Partnerships in AoW 3 include the Secretariats of the Plant Treaty, CBD, CGRFA, National Plant Treaty Focal Points, national genebanks, DSI Scientific Network, Art 15 DGs and System Board, International genebanks, and Breeding for Tomorrow.

AoW 4 (Germplasm Health) will ensure compliance with national quarantine laws as materials move across international borders. AoW 4 will develop and improve phytosanitary interventions to meet current and emerging phytosanitary demands, and identify, develop, and mainstream effective, advanced phytosanitation and diagnostics technologies (genomic-based molecular methods, multispectral imaging, AI/bioinformatics, etc.) for routine germplasm health monitoring. Partnerships in AoW 4 include Sustainable Farming and Breeding for Tomorrow, FAO-IPPC, NPPOs, RPPOs, Universities, ISTA, ARIs, and NARES.

AoW 5 will contribute to more effective, integrated ways of conserving and using diversity worldwide, enhancing the PGRFA Global System. Capacity needs relating to plant genetic resources conservation and use are expressed in detail by regions, countries, and institutions in submissions to the Plant Treaty Governing Body, FAO State of the World reports, reports on the Global Plan of Action, crop strategies and genebank reviews undertaken by the Crop Trust, and reviews of national breeding capacity undertaken by the Accelerated Breeding Initiative. There is tremendous demand for increased capacity sharing and strengthened mechanisms for partnership and networking from CGIAR genebanks and GHUs. CGIAR Centers have an expansive reach for supporting countries worldwide to conserve and use their genetic resources and ensure national breeding programs and diverse users benefit. Germplasm acquisition, conservation and distribution, together with capacity-sharing activities will benefit thousands of genebank managers and their teams, and users and beneficiaries annually, who in turn will better conserve and distribute germplasm and share knowledge with thousands of their own users. The genebanks, GHUs and policy teams will share their expertise and capacity, gained in the other AoWs, with regional and national partners to connect and integrate collections on a global scale and create a more vibrant and dynamic two-way flow of germplasm between in situ and ex situ. AoW 5 partnerships include national genebanks, Crop Trust NARES, NPPOs, NGOs, Civil Society Organizations, and Open University, Food Frontiers and Security and CapSha.

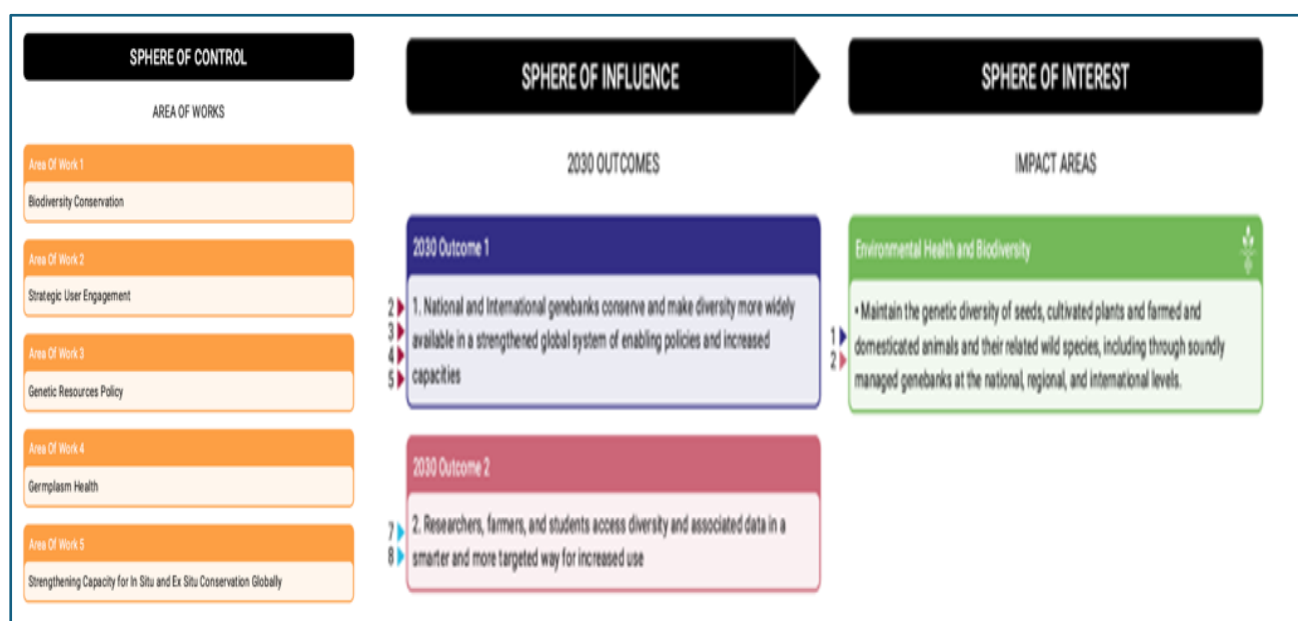
Genebanks activities will contribute to CGIAR Impact Area (5) Environmental health and biodiversity and the Collective Target 11 By conserving and making available biodiversity, Genebanks are responding to SDG 2.5 and the Kunming-Montreal Global Biodiversity Framework

Targets 4, 10, 13, and 20, and the FAO's Global Plan of Action on Plant Genetic Resources for Food and Agriculture (PGRFA) and giving thousands of users access to plant diversity sources. Genebanks will indirectly contribute towards CGIAR Impact Areas (1), (2), (3) and (4).

Full ToC Diagram

Genebanks developed Objectives and Research Questions that are found in the Area of Work narratives.

OBJECTIVES: Genebanks aims to ensure the reliability of conservation and genebank services now and into the future and to build a stronger global system of policies, mechanisms, institutions and conservationists making genetic diversity more widely



Results framework

Program/Accelerator results framework

Please refer to the Results Framework attachment

2. Monitoring plan for HLOs and outcomes

The genebanks will continue to use an online reporting tool (ORT), which was set up in 2014 to monitor the status of individual collections with respect to performance targets. Through the QMS adopted in AoW 1, genebank operations, data, and accession numbers are audited internally and externally and are subject to regular review by the Crop Trust. The audits ensure that international standards are complied with, and deviations and risks are documented and managed. FAO's monitoring of UN SDG 2.5, State of the World of PGRFA, and the Global Plan of Action, the Crop Trust's oversight of long-term partnership agreements and grants, CBD's Global Biodiversity Framework, and perhaps most importantly, CGIAR Centers' Article 15 Agreements with the Plant Treaty's Governing Body all require annual, biannual, or periodic submissions from CGIAR and help monitor outcomes of CGIAR genebanks in all five AoWs. In addition, individual accessions, their data, and the availability status of national and international genebanks are also recorded and regularly updated in the Crop Trust's online portal, Genesys.

AoW and communities of practice for seed and clonal crops and data management will be coordinated by dedicated staff, who will be responsible for implementing workplans and collating information and feedback from each genebank and GHU. Progress on high-level outputs will be reported also in the ORT in a suitable format to comply with system-level annual reporting requirements. By nature, AoW 2 will involve an iterative process of identifying target user typologies to engage and develop user entry points into genebank data resources. The success of this AoW will become evident in trends in the use of the online portal and germplasm requests monitored under AoW 1. Genebanks will adopt a robust feedback mechanism to monitor rates of response to users' requests and their satisfaction. Bilateral work relates to research projects, which will not be monitored through Genebanks. However, the results of such work will provide new data and information to aid the selection of genebank materials, which will become evident in trends in germplasm requests and the use of materials.

Data collection strategy

Please refer to the Results Framework attachment

The ten CGIAR genebanks within the Accelerator will report in ORT annually. The ORT reporting is done in a format to comply with system-level annual reporting requirements, including but not limited to the submission of evidences of results, providing progress against planned results and targets, and the description of challenges and achievements. Dedicated staff members, which include the Leader, MELIA officer, and Coordinator, review the ORT reporting before reporting in the PRMS.

Standard Operating Procedures (SOPs) define how conservation data is captured and recorded. It varies from data type and genebank. Usually, raw data is stored and later curated by the specific team, revised by the curator and finally approved by the genebank head to be updated to the system as final data. There are barcodes and systems that support the process of being more efficient and reducing potential mistakes. All data is also backed up either by the genebank itself on a different server and/or by the Center IT Unit. Data quality assurance is the role of the curators.

3. Evaluation plan

Evaluation studies

Please refer to table 1. MELIA studies in Annex 1.

Evaluation questions include:

- How did the Seed Quality Management (SQM) and Quality Management System (QMS) Communities of Practices improve the effectiveness and efficiency of CGIAR genebanks? (AoW 1)
- To what extent are the partners that have received training from CGIAR genebanks applying the learnings and what are other critical areas that require further focused support? (AoW 5)

Evaluation collaboration plans

Genebanks will seek to collaborate with the Multifunctional Landscapes Program to study in situ and on farm site monitoring of diversity conservation. Genebanks will also collaborate with the Crop Trust and Aarhus University to review the Scientific Impact of the Seed Quality Management and Quality Management System Communities of Practice. Collaborations with W3/bilateral projects will be explored during phase 2 of the mapping exercise.

4. Impact assessment

Impact assessment (IA) is assessment that estimates the causal effects of research outputs and related activities and assesses the magnitude of impact achieved. Impact assessments are conducted after an intervention has been completed and are used to substantiate claims of sustained results. They seek to measure impact at scale, i.e., benefits to a large number of people/large areas of land. It employs both quantitative and qualitative methods, often combined for deeper insights.

IA goes beyond monitoring indicators to evidenced attribution of observed changes to CGIAR interventions. IA will help to ensure that Programs/Accelerators contribute meaningfully to [CGIAR's Results Framework](#) and the 2025–30 Portfolio objectives.

Program/Accelerator impact overview

Genebanks, through five interconnected Areas of Work, will enable diversity conservation in a more efficient and effective manner (AoW 1), smarter and more targeted access to diversity for increased use (AoW 2), and a supportive policy environment (AoW 3) while controlling phytosanitary risks (AoW 4). The genebanks, GHus, and policy teams will share their expertise and capacity with regional and national partners to connect and integrate collections on a global scale and create a more vibrant and dynamic two-way flow between in situ and ex situ (AoW 5).

Genebanks activities will lead to a stronger overall global system of international and national institutions, policies, and mechanisms contributing to the conservation of plant genetic resources, which contributes to CGIAR Impact Area (5) Environmental health and biodiversity and the Collective Target “Maintain the genetic diversity of seed varieties, cultivated plants and farmed and domesticated animals and their related wild species, including through soundly managed genebanks at the national, regional, and international levels”. By conserving and making available biodiversity, Genebanks are responding to SDG 2.5 and the Kunming-Montreal Global Biodiversity Framework Targets 4, 10, 13, and 20, and the FAO’s Global Plan of Action on Plant Genetic Resources for Food and Agriculture (PGRFA) and giving thousands of new users access to plant diversity sources. Genebanks will indirectly contribute towards CGIAR Impact Areas (1), (2), (3) and (4).

The impact pathway of the germplasm disseminated annually from genebanks is not easily tracked. It is notoriously difficult to keep track of materials as they pass through phases of evaluation, selection, crossing, or passing on to other researchers and national partners. Although all CGIAR genebanks have implemented digital object identifiers (DOI) minted by the Plant Treaty, so far, few users have adopted the same DOI. Ideally, this would change, and standards developed between Genebanks and Breeding for Tomorrow would improve sample tracking and data gathering and could serve as an example to be followed by other scientists within and outside the CGIAR. A few pedigree tracing studies have been carried out and indicate that genebank materials have introduced important traits and genetic diversity into breeding lines, contributing to genetic gain (Villanueva et al.¹, 2020, Sellitti et al.², 2020, Bernal-Galeano et al., 2020³).

In general, genebanks respond to requests and do not direct where materials are sent or what they are used for. Subject to ‘direct use’ exceptions discussed below, the genebanks are often upstream providers of services, introducing genetic variation into a potentially long breeding pipeline, with impact only measurable after several years. In this context, Genebanks aim is to ensure that genebanks and GHUs provide good quality services and offer genetic variation of value. In addition, Genebanks will strive to adopt a more dynamic curatorial approach to

¹ Villanueva et al. (2020) The contribution of the International Rice Genebank to varietal improvement and crop productivity in Eastern India. *Food Security*, 12: 929–943

² Sellitti et al. (2020) The contribution of the CIAT genebank to the development of iron-biofortified bean varieties and well-being of farm households in Rwanda. *Food Security*, 12: 975–991

³ Bernal-Galeano, V. et al. (2020). Andean potato diversity conserved in the International Potato Center genebank helps develop agriculture in Uganda: the example of the variety ‘Victoria’. *Food Security*. ISSN 1876-4525. Published online 22May2020.

structuring collections, which will enable conservation of specialist collections and genetic stocks on a temporary basis. Finding cost-efficient solutions to testing and cleaning clonal crop materials is a shared priority for breeders, GHUs, and genebanks, and it will be beneficial if tackled in a collective approach.

The use of genebanks and genetic diversity can be increased through the improvement of data and screening tools and undertaking trait discovery studies, such as (environmental) genome-wide association studies or digital sequencing, which help to speed up the process of identifying useful traits and genetic variation of relevance for target product profiles. AI approaches promise to facilitate more effective and large-scale screening of genetic diversity for new traits. Diversity studies for different crops have repeatedly illustrated that current breeding efforts have only tapped into a small part of the genetic diversity available in the cultivated crop gene pools, let alone the diversity represented in wild relatives. A considerable genetic reservoir of potential solutions lies untapped in genebanks, which may be made more accessible through enriched data and germplasm identification mechanisms. Genebanks will amalgamate data as well as seeds and partner with researchers to enrich and manage data resources in a harmonized way.

In addition, there is growing evidence of demand for materials from genebanks for ‘direct use in production’. This is the case for example with some crops with less well resourced plant breeding programs, and where there is traditionally a much shorter impact pathway to the ultimate beneficiaries. This is the case for crops of regional or localized importance (e.g., ulluco, oca, winged bean, bambara groundnut, jute mallow, barnyard millet); roots, tubers and bananas, for which virus-clean material contributes directly to increased yields; varieties for niche uses or specialized markets; forages; trees; heritage varieties repatriated to communities; crops for which breeding is challenging or which have been historically under resourced (e.g., amaranth, okra) and crops in which farmers’ varieties have important applications in specific environments (e.g., banana, yams, cassava, potato, sweetpotato, yacón). There is also evidence that the movement of farmers’ varieties with specific climatic or biotic adaptations to new geographical locations experiencing climate change or disease pressure has significant benefits, as does introducing genetic diversity from crop Centers of origin across continents to geographies where diversity is lacking (e.g., cassava in Asia). Furthermore, even in major crops such as rice there is sufficient potential for direct-use as evidenced by the results in WP4 in Genebanks Initiative (2022-24). In all these instances, the genebanks have not articulated or elaborated where their contributions may have high returns on investment.

Evidence from studies such as Luettringhaus et al. (2021)⁴ and Molina et al. (2022)⁵ highlights gender, age, and socio-economic differences in germplasm management on farm, yet significant gaps remain regarding the impact of genebank services on women’s livelihoods and empowerment. AoW 2 (Strategic User Engagement) offers a key opportunity to close these gaps by developing inclusive strategies that recognize gendered roles and barriers, integrate women’s knowledge and priorities, and create targeted pathways for women, youth, and marginalized

⁴ Lüttringhaus, S. et al. (2021). "Dynamic guardianship of potato landraces by Andean communities and the genebank of the International Potato Center." *CABI agriculture and bioscience* 2: 1-16.

⁵ Molina, C.A. et al. (2022). Gender Roles and Native Potato Diversity Management in Highland Communities of Peru. *Sustainability*, 14, 3455. <https://doi.org/10.3390/su14063455>

groups to access, use, and benefit from genebank resources. AoW 3 explores policies to increase the flow of such benefits to farmers, women, youth, indigenous peoples and local communities from commercial users of PGRFA.

In a more focused way, through AoW 5, individual genebanks will connect with communities on a small scale to encourage a two-way flow of germplasm and help to build capacity and provide safety backups to support in situ conservation of crops of mutual interest. Several genebanks nurture existing in situ – ex situ connections with communities where there remains a strong cultural connection to crop diversity and traditional knowledge. The custodians of such diversity are mostly from an older generation who have actively conserved their agricultural heritage against strong pressures to change. However, they struggle to interest younger generations to do the same. It is evident that without additional support these cultures, varieties and knowledge will follow a path to extinction that many other crops and cultures have previously trodden. In countries where they have a strong presence, international genebanks may have the taxonomic expertise, legal status, and reliable conservation practices to work with such marginalized, diversity-rich communities to provide support in various ways and ensure that they can improve the way they conserve their diversity in situ and at the same time access genebank resources, so neither genetic diversity nor traditional knowledge are lost.

IA activities and collaboration plans

The impact of genebanks is notoriously challenging to assess given the lack of germplasm tracking, obscure pedigrees of improved materials, complex attribution patterns along a long impact pathway, and the wide range of users and purposes to which genebank materials are put. Much has been published on the economic impact of traits or diversity sourced from genebanks, and some cost-benefit analyses have concluded that the costs of long-term conservation will be minor compared to the scale of impact and returns on investment of just one disease-resistant trait coming from screening a genebank collection in a major crop. One of the most impressive impacts occur where “black swan” events (e.g., UG99 wheat stem rust, banana Fusarium tropical race 4) demand the screening and use of diversity found nowhere else but genebanks. Opportunities will be taken to participate in large, rigorous studies undertaken by downstream partners and Science Programs to which genebanks contribute. For instance, Genebanks may benefit from adoption studies undertaken by Breeding for Tomorrow or other studies from different breeding programs. AoW 2 will improve germplasm tracking, its use, and follow up with users to monitor potential outcomes of ordering genebank materials. The increasing use of common standard DOIs by CGIAR and other breeders would considerably improve tracking the contribution of genebank materials to widely adopted improved varieties. Impact stories have been written by postgraduate students and have highlighted the contributions of genebanks to varieties growing in farmers’ fields. While these published studies are limited in depth and scope, they suit the scope of genebanks as service providers and the wide range of impacts to which they contribute.

IA studies

Please refer to table 1 in Annex 1.

AoW 4 will develop an Impact Assessment study with the tentative title “From Exchange to Impact: How GHUs Catalyze Genebanks-Breeding Germplasm Exchange for Invasive Cassava Disease Management” in collaboration with Breeding For Tomorrow, Sustainable Farming, IITA, and partners. The geographic scope will be Nigeria, Thailand, and Vietnam. The study will seek to validate the assumption “Enabling phytosanitary procedures and policies for import and export.”

5. Learning and adaptive management

Learning

The Quality Management System (QMS) adopted in AoW 1 (Biodiversity Conservation) allows genebanks to document new and regularly update existing standard operating procedures (SOPs), complying with evolving international standards (FAO, 2014). The approach enables continuous improvement, exchange of best practices, and harmonization of terminology and processes across Centers. Genebank staff across Centers are responsible for specific processes convene in Process Teams, which meet regularly and provide an enabling mechanism for mainstreaming new practices and sharing experiences. Members of the Process Teams will self-organize to conduct peer-to-peer audits of each other’s SOPs. Risk management is a requirement of the genebank QMS and is critical to responding to a wide range of events. Genebank operations, data, and accession numbers are audited internally and externally and are subject to regular review by the Crop Trust. External reviews held by the Crop Trust come up with recommendations for each genebank, identifying the specific and critical issues that need to be addressed. The MELIA study to review the scientific impact of the SQM and QMS Communities of Practice, in line with Genebanks Platform Evaluation findings, will review the contribution of the CoPs to the effectiveness and efficiency of genebanks, to learn lessons from the successes of the CoPs.

Areas of Work and communities of practice for seed and clonal crops and data management will be coordinated by dedicated staff, who will be responsible for implementing workplans and collating information and feedback from each genebank and GHU. By nature, the Strategic User Engagement AoW will involve an iterative process of identifying target user typologies to engage and develop user entry points into genebank data resources. The success of this Area of Work will become evident in trends in the use of the online portal and germplasm requests monitored under AoW 1 (Biodiversity Conservation). Genebanks will adopt a robust feedback mechanism to monitor rates of response to users’ requests and their satisfaction.

Genebanks Management team meets on a monthly basis to share financial, technical, and management updates. Discussions held during the meeting support better interaction and co-learning among genebanks, policy, GHUs and management, promote better interaction, and proper follow up on achievements, guiding decision making and potential adjustments to the Accelerator.

The Annual Genebank Meetings (AGMs) are an important forum to review and reflect on the CGIAR Genebanks past year's activities, progress and achievements, and to make plans for the next year. National partners, CGIAR programs, and technical and scientific experts are represented in the AGMs, and the location of the meeting is in a different region each year to assure better interaction with multiple stakeholders from different regions. By doing this, we are constantly consulting with partners in different regions of the world, understanding their realities and needs so we can better plan the priorities based on a face-to-face consultation with partners. Topics discussed at the AGMs are taken forward to the management team meeting for further discussion and action. From these consultations with multiple partners, we have developed the structure of the capacity sharing activities (such as webinars, workshops, and training courses) targeted within and outside genebanks.

Annual review and adaptive planning

The Program/Accelerator will build on the experience of the 2022–24 Portfolio approach to adaptive management. Once updated guidance on the adaptive management is available, details key changes made to delivery plans and the rationale for those changes will be documented and tracked.

During the 2022-2024 cycle, the CGIAR Initiative on Genebanks participated in the Report-Reflect-Replan annual adaptive management cycle, as part of the CGIAR Technical Reporting Arrangement. The Reflect process each year allowed the Initiative to assess its implementation in the previous year and make recommendations for how to adapt and respond to lessons learnt which were included in the Initiative 2022 and 2023 Annual Technical Reports (Type 1) (Section 7 – Adaptive Management). The Genebanks Accelerator builds directly on the CGIAR Genebanks Initiative (2022-2024), and previously the CGIAR Genebank Platform (2017–2021) and the Genebank CGIAR Research Program (2012–2016). Genebanks will continue to annually reflect, adapt and respond to changes in budgets and circumstances, and to lessons learned in line with updated guidance once available.

6. MELIA resources

MELIA budget and staff

Please refer to table 2 in Annex 2.

7. Data management

General approach

Genebanks will continue to use an online reporting tool (ORT) to monitor the status of individual collections with respect to performance targets. In addition, individual accessions, their data, and the availability status of national and international genebanks are also recorded and regularly updated in the Crop Trust's online portal, Genesys. Progress on high-level outputs will be reported also in the Online Reporting Tool in a suitable format to comply with system-level annual reporting requirements. Genebanks will continue using Sharepoint to collect, use, manage, and store all documentation related to active genebank operations. The databases (such as dataverse and GIGWA) used by the Genebanks for data management, storage, and analysis are compliant with the [CGIAR Open and FAIR Data Assets Policy](#).

Annex 1. MELIA studies

Table 1. MELIA studies

MELIA Study	Lead Center and contact	Result level	Result(s) evidenced by this study	Indicator(s) evidenced by this study	Assumption(s) assessed by this study	Geographies (scope)	Methods and design approaches	Other Program(s)/ Accelerator(s)	Start date for the study	End date for the study	Study cost (USD)	How related to past evaluations (SG or others)
Assessing the application of best practices among national partners	AfricaRice	Intermediate Outcome	Partners apply best practices to the management and exchange of agrobiodiversity	Number of national partners developing new protocols		Africa	Survey sent to trainees Selection of partners for assessment visit. Conduct assessment visit in two countries in 2025 Develop a recommendations report post visit		2025	2030	TBD	
In situ and on farm site monitoring of diversity conservation including gender aspects	AfricaRice and ICARDA	Intermediate Outcome	Partners apply best practices to the management and exchange of agrobiodiversity	Number of sites monitored for in-situ / on farm		WEST AFRICA and CWANA	In 2025, the methodology will be developed including the identification of sites to be monitored for in-situ and on farm for wild relatives and landraces	Multifunctional Landscapes	2025	2030	TBD	
Study on the Use of Germplasm	IRRI	2030 Outcome	Researchers, farmers, and students	Change in user requests		Global	In 2025, the baseline study will be		2025	2030	TBD	The 2023 Genebanks Evaluations

Genebank Accelerator MELIA Plan

MELIA Study	Lead Center and contact	Result level	Result(s) evidenced by this study	Indicator(s) evidenced by this study	Assumption(s) assessed by this study	Geographies (scope)	Methods and design approaches	Other Program(s)/ Accelerator(s)	Start date for the study	End date for the study	Study cost (USD)	How related to past evaluations (SG or others)
Requested from CGIAR Genebanks			access diversity and associated data in a smarter and more targeted way for increased use	to crop diversity			developed through document review, surveys, and data analysis. The baseline study will be followed by an outcome study in the next 5 years.					Recommended to-Adopt a wider definition of users when designing outreach programs and to expand efforts to encourage and facilitate the use of genebank collections.
Review the Scientific Impact of the Seed Quality Management (SQM) and Quality Management System (QMS) Communities of Practice	CIAT	Intermediate Outcome HLO	CGIAR genebanks conserve and make available crop diversity in perpetuity with greater efficiency and effectiveness HLO 1.1.2. Innovations to enhance the efficiency of conserving collections			Global	Review of reports Interviews		2025	2026	TBD	The 2023 Genebanks Evaluations Recommended to Review Scientific Impact of the Communities of Practice - they represent an innovative approach which may be an exemplary model for organizations beyond CGIAR

Genebank Accelerator MELIA Plan

MELIA Study	Lead Center and contact	Result level	Result(s) evidenced by this study	Indicator(s) evidenced by this study	Assumption(s) assessed by this study	Geographies (scope)	Methods and design approaches	Other Program(s)/ Accelerator(s)	Start date for the study	End date for the study	Study cost (USD)	How related to past evaluations (SG or others)
From Exchange to Impact: How GHUs Catalyze Genebanks- Breeding Germplasm Exchange for Invasive Cassava Disease Management	IITA	Intermediate Outcome	CGIAR and internal and external partners exchange genetic resources without the risk of spreading quarantine pests or diseases		Enabling phytosanitary procedures and policies for import and export.	Nigeria, Vietnam, Thailand	In 2025, the concept and methodology will be developed in addition to the draft ToR for the consultant to be employed on this case study in 2026 for 10 months duration	Breeding For Tomorrow, Sustainable Farming	2025	2026	TBD	
Harmonization of the post distribution surveys across genebanks	ICARDA	Intermediate Outcome	CGIAR genebanks conserve and make available crop diversity in perpetuity with greater efficiency and effectiveness				Review of reports		2025	2025	TBD	

Annex 2. MELIA budget

Genebanks allocated 50K to conduct MELIA studies in 2025 (under AoW0 ICARDA). The table below doesn't show the full budget since the costs are still being estimated.

Table 2. MELIA Budget: Staff Cost Breakdown in 2025.

Activity title	Activity description	Staff (# per type)	Full-time equivalent (# days)	Staff cost (USD)	Inputs i.e. travel costs, special equipment costs	Inputs cost (USD)	Total cost (USD)
Monitoring activities: Cross-cutting Activities related to the routine monitoring of HLOs and outcomes							
Refine the Online Reporting Tool	Refine the Online Reporting Tool in a suitable format to comply with system-level annual reporting requirements	1 MELIA staff 1 Coordinator 1 Consultant	5 days 5 days 20 days	TBD	NA	NA	TBD
MELIA studies: Cross-cutting MELIA studies to evidence the achievement of program-level impacts and outcomes							
Study on the Use of Germplasm Requested from CGIAR Genebanks	Although the SMTA of MLS of the Plant Treaty states that there is no need to track the use of the germplasm, it is highly important to understand how the germplasm distributed by the CGIAR have been impacting agriculture, especially in the global south. Although the number of samples distributed is enormous, very few impact studies have been conducted so far. The baseline study in 2025 aims to look into the use of material in the past years through document review and user surveys that will be followed	1 MELIA officer 2 Genebank staff 1 Consultant	10 days 4 days 90 days	TBD 15 K	NA	NA	TBD 15 K

Genebank Accelerator MELIA Plan

Activity title	Activity description	Staff (# per type)	Full-time equivalent (# days)	Staff cost (USD)	Inputs i.e. travel costs, special equipment costs	Inputs cost (USD)	Total cost (USD)
	by an outcome study after the launch of the online portal.						
MELIA studies: AoW outcomes							
MELIA studies to evidence the achievement of AoW-level intermediate outcomes not covered by any cross-cutting MELIA studies							
Assessing the application of best practices among national partners	Genebanks will follow up with partners that have received training to evaluate the implementation of the learnings and provide further support focused on critical areas that may still require further support. In 2025, the study will target assessing the application of best practices by selected national partners in two other countries.	MELIA officer	5 days	TBD	NA	NA	NA
		2 Genebank staff covering country A	10 days		Travel costs	10 K	10 K
		2 Genebank staff covering country A	10 days		Travel costs	10 K	10 K
In situ and on farm site monitoring of diversity conservation including gender aspects	Genebanks will seek to collaborate with Multifunctional Landscape Science Program to co-learn and co-develop harmonized strategies for in situ monitoring that can be summed (from both programs) and analyzed jointly, promoting a more impactful result of the CGIAR Portfolio, due to the possibility of covering more countries and in situ/on farm communities. In 2025 the methodology will be developed including the identification of sites to be monitored for in-situ and on farm for wild relatives and landraces.	1 MELIA officer 2 Genebank staff from two Centers 1 scientist from Multifunctional Landscapes	5 days 3 days 3 days	TBD	NA	NA	TBD
Harmonization of the post distribution surveys across genebanks	Individual genebanks send requestors post distribution surveys aiming at measuring user satisfaction and other factors. The study aims developing a	1 MELIA officer	5 days	TBD	NA	NA	TBD

Genebank Accelerator MELIA Plan

Activity title	Activity description	Staff (# per type)	Full-time equivalent (# days)	Staff cost (USD)	Inputs i.e. travel costs, special equipment costs	Inputs cost (USD)	Total cost (USD)
	common survey among the genebanks						
Review the Scientific Impact of the Seed Quality Management (SQM) and Quality Management System (QMS) Communities of Practice	The study aims to measure the efficiency of the SQM and QMS CoPs within Genebanks. It will compare the conditions before and after the CoPs and look into specific cases.	1 MELIA officer 1 Consultant (Crop Trust) 1 Consultant (SQM)	20 days 10 days 5 days	TBD	NA	NA	NA
Impact Assessment: Enabling germplasm exchange between Africa and Asia contributes to disease management	In 2025, the concept note and methodology will be developed in addition to the draft ToR for the consultant to be employed on this case study in 2026 for 10 months duration	1 GHU scientist	5 days	TBD	NA	NA	TBD
Total MELIA activity costs				TBD		TBD	TBD

